AXI4-Lite Slave IP with Memory

Pre-Silicon Verification with UVM

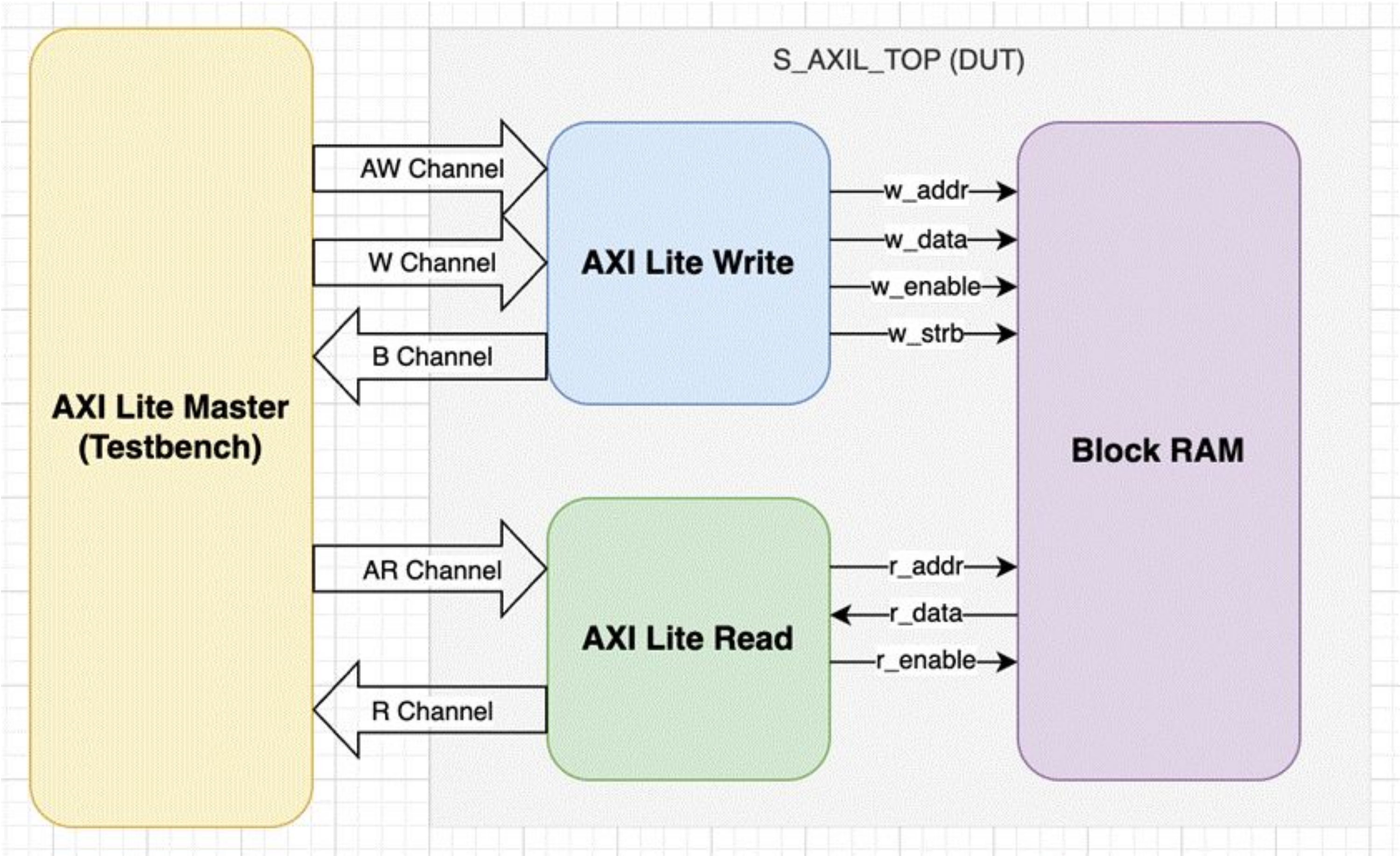
Milestone 5 — Final Submission

ECE-593: Fundamentals of Pre-Silicon Validation
Instructor: Venkatesh Patil
Portland State University | Spring 2026

Team1: Patrick + Pratibha

May 28, 2026

IP Overview — Block Diagram



Module Hierarchy

```
s_axil_top
  ├─ s_axil_wr
  ├─ s_axil_rd
  └─ mem[256]
```

Key Parameters

DATA_WIDTH = 32

ADDR_WIDTH = 12

MEM_DEPTH = 256 (1 KB)

MEM_INIT = hex preload file

IP Functional Description

Write Path

- AW and W channels accepted independently
- Write fires when both channels latched & no B pending
- WSTRB controls byte-lane masking
- BRESP = OKAY in-range, SLVERR out-of-range (dropped)

Read Path

- AR channel accepted; memory read issued next cycle
- Data presented on R channel one cycle later
- RRESP = OKAY in-range; out-of-range returns 0x0 + SLVERR
- At most one outstanding read at a time

Reset Behaviour

- Active-low synchronous resetn clears all control state
- Memory contents not reset; optionally preloaded via MEM_INIT

Write-Before-Read Forwarding

- Write and read to same address in same cycle
- Incoming write data forwarded directly to read output
- Avoids stale-read window; WSTRB applied during forward

Response Codes

- **OKAY (2’b00)** — in-range read or write
- **SLVERR (2’b10)** — address ≥ MEM_DEPTH

AXI4-Lite Handshake

- All 5 channels use standard VALID/READY protocol
- READY signals are combinational (no pipeline stalls)
- Single outstanding transaction per channel

Design Under Verification & UVM Environment

Design Under Verification

A parameterized AXI4-Lite slave wrapping a 256-word × 32-bit block RAM. DATA_WIDTH = 32, ADDR_WIDTH = 12, MEM_DEPTH = 256. Supports byte-strobe (WSTRB) write masking, write-before-read forwarding, and two AXI response codes — OKAY for in-range access and SLVERR for out-of-range. Out-of-range writes are dropped; out-of-range reads return RDATA = 0. Single outstanding transaction; ready signals are pure combinational functions of the per-channel active flags.

UVM Environment

Standard UVM 1.2 environment under Synopsys VCS R-2020.12: **axil_test** instantiates **axil_env**, which contains the **axil_agent** (sequencer / driver / monitor) and the **axil_scoreboard**. The monitor analysis port feeds the scoreboard, which maintains a WSTRB-aware shadow-memory model and runs both transaction-level and cycle-level covergroups.

Six sequence classes drive stimulus: random, write/read-same-address, byte-strobe, concurrent write/read, out-of-range, and unaligned address. The scoreboard's **check_phase** guarantees that a non-zero number of transactions was observed by end of test, eliminating the silent zero-transaction PASS condition.

Verification Approach

Stimulus

A 50-seed constrained-random regression provides the bulk of stimulus, augmented by six directed sequences that target specific scenarios: byte-strobe patterns, concurrent read/write to the same address, out-of-range accesses (forces SLVERR), and unaligned addresses (toggles the low address bits). All sequences are parameterised on DATA_WIDTH, ADDR_WIDTH, and MEM_DEPTH and run through the standard UVM sequencer.

Checking

The scoreboard maintains a shadow-memory model that mirrors the DUT, applying WSTRB on writes and comparing every read against the predicted value. Response codes are checked against the in-range / out-of-range policy. A check_phase asserts non-zero observed transactions at end of test, so a structural disconnection between monitor and scoreboard cannot produce a silent PASS.

Functional Coverage

Coverage Group	Bins / Cross Points	Achieved
cg_reset	Reset asserted / released states	100%
cg_aw_w_hs	AW/W handshake order: AW-first, W-first, concurrent	100%
cg_bresp	BRESP ∈ {OKAY, SLVERR} × BREADY delay bins	100%
cg_wstrb	WSTRB values 0001 … 1111 (15 non-zero patterns)	100%
cg_oor	Out-of-range addr × op {WR, RD} × response	100%
cg_ar_hs	AR-to-R latency bins {1, 2, 3, 4-7, 8-15, 16+}	100%
cg_rresp	RRESP ∈ {OKAY, SLVERR}	100%
cg_fwd	WR + RD same cycle × address match (forwarding)	100%
cg_addr	Word addresses binned across [0, MEM_DEPTH-1]	100%
cg_bp_b/cg_bp_r	BREADY/RREADY stall bins {0, 1-3, 4-7, 8-15, 16+}	100%/100%
cg_b2b	Back-to-back WR-WR, RD-RD, WR-RD, RD-WR pairs	100%

Code Coverage

Code coverage: **96.72%** Functional coverage **100%**. The remaining condition and toggle gap was reviewed and waived as architecturally unreachable.

Metric	Achieved
Line	100.00 %
Branch	94.29 %
Functional	100.00 %
Toggle	98.74 %
Condition	90.57 %
FSM	N/A
Overall	**96.72 %**

- **Toggle — bresp[0] / rresp[0] / b_resp_lat[0] / r_resp[0]**
  - **Only have 2 response codes in our RTL**
- **Branch — MEM_INIT == "" (Lines 73 & 80 of s_axil_top)**
  - **Testbench always init with a file**
- **Condition — awvalid=1 while awready=0 (Line 85) and wvalid=1 while wready=0 (Line 105)**
  - **Valid signals are driven by driver after seeing ready**
- **Condition — arvalid=1 while arready=0 (Line 77)**
  - **Same as above**

Bug Injection — Two Bugs, Two Catches

Two bugs were injected to demonstrate that the environment actively detects failures. Both were reverted after demonstration.

1. RTL bug — WSTRB byte-lane masking defeated

Mutation: in s_axil_wr.sv, `assign mem_wstrb = w_strb_lat;` replaced with `assign mem_wstrb = '1;` — memory writes all four byte lanes regardless of WSTRB.

Expectation: scoreboard's WSTRB-aware shadow memory diverges from the DUT on any partial-strobe write; mismatch should fire on the next read.

Result: caught. 261 UVM_ERROR `[SCB] READ MISMATCH` across the regression; scoreboard reports FAIL. Mismatch signature confirms WSTRB was ignored.

2. Environment bug — monitor analysis port disconnected

Mutation: in axil_env.sv connect_phase, the line `agent.monitor_port.connect(scoreboard.analysis_imp);` is commented out — monitor sees traffic, scoreboard never receives it.

Expectation: scoreboard writes_checked / reads_checked stay at zero; the new `check_phase` should fire UVM_ERROR for zero observed transactions.

Result: caught. UVM_ERROR `[SCB] ENV CHECK FAILED` fired; scoreboard reports FAIL (env check). Before this milestone added check_phase, the same bug would have silently reported PASS.

Takeaways

Unreachable analysis is part of the deliverable

The residual ~3% coverage gap was reviewed line by line. Combinational READY signals make VALID-while-not-READY architecturally impossible; BRESP/RRESP bit 0 is constant by protocol; FSM is not applicable because the control logic uses independent handshake flag registers rather than an encoded state machine. Each gap was waived with rationale rather than left silent.

Self-improving verification

Injecting the environment bug exposed a gap we had not anticipated: with zero transactions observed, the scoreboard would have silently reported PASS. The fix — a scoreboard `check_phase` that asserts a non-zero transaction count — is now a permanent guard against that failure mode for every future regression. This is verification working on the verification environment, and the most important takeaway from MS5.